CMSC 137

Data Communications and Networking

Lecture Module

05 - Signal

University of the Philippines
LOS BAÑOS

Module 5: Signal

Rozano S. Maniaol
Lecturer

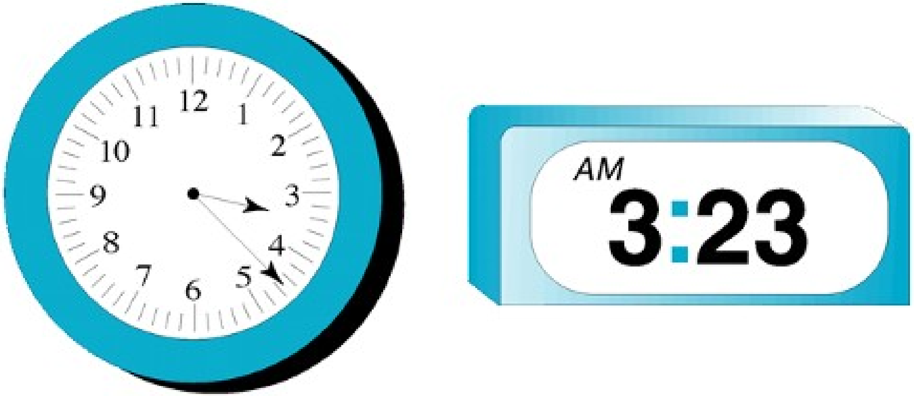
Introduction

Physical layer moves data in the form of electromagnetic signals across a transmission medium

Data must be changed to a form that transmission media can accept

Data

- Analog data
  - Continuous, e.g human voice
- Digital data
  - Discrete state, e.g data produced by computers

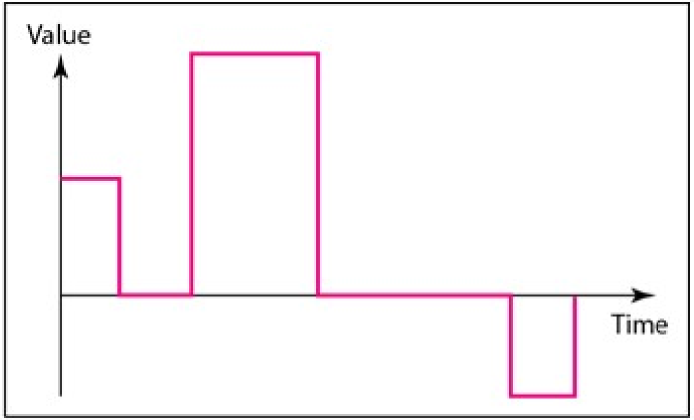


a. Analog b. Digital

Signal

- Analog signal
  - Infinitely many levels of intensity over a period of time
- Digital signal
  - Have a limited number of defined values (ex. 0 and 1)

a. Analog signal

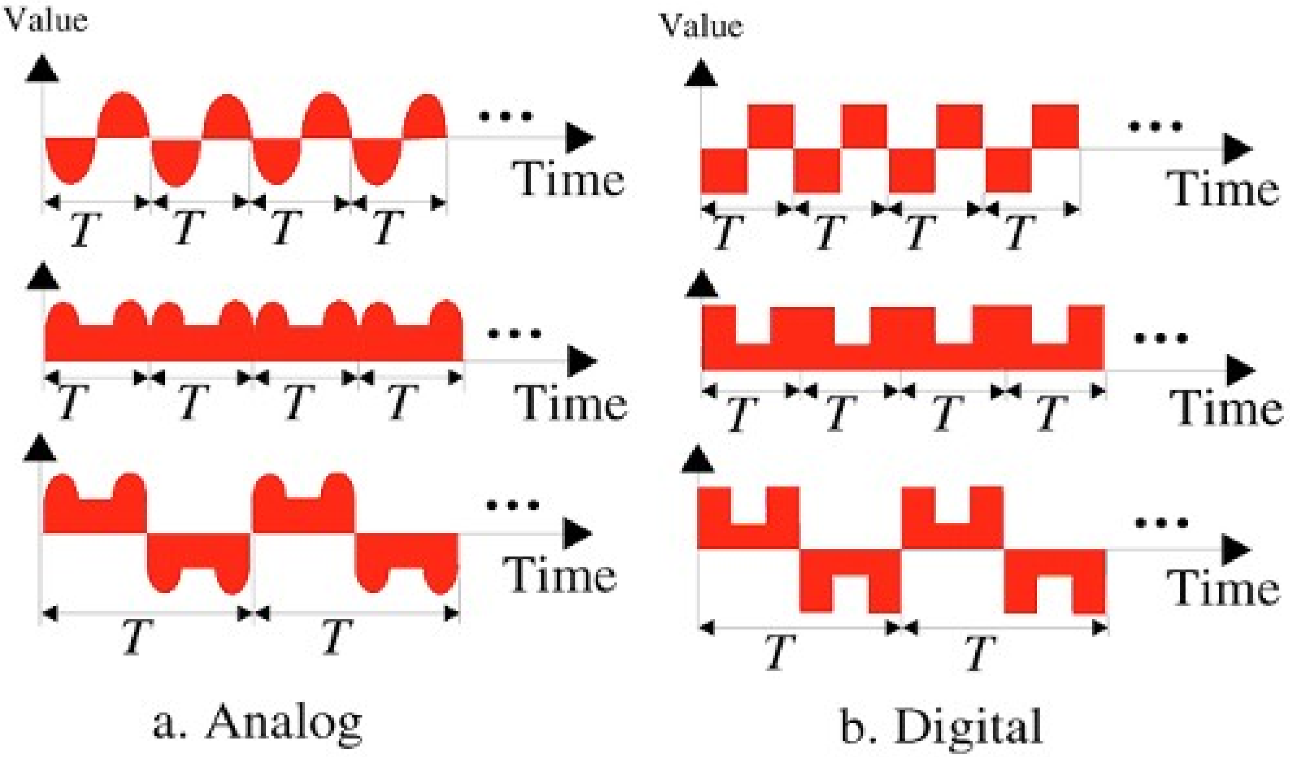


b. Digital signal

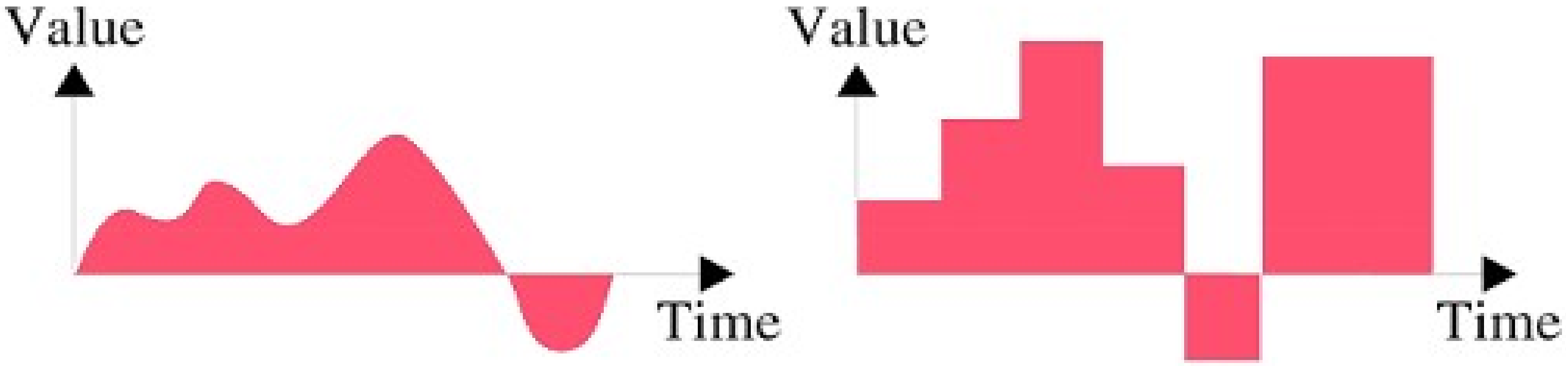
Periodic and Nonperiodic Signal

- Periodic signal
  - Completes a pattern within a measurable time frame (period)
  - Cycle – completion of one full pattern
  - Need less bandwidth
- Nonperiodic signal (Aperiodic)
  - Changes without exhibiting a pattern or cycle that repeats over time
  - Can represent variation in data

Periodic Signals



Nonperiodic Signal



a. Analog

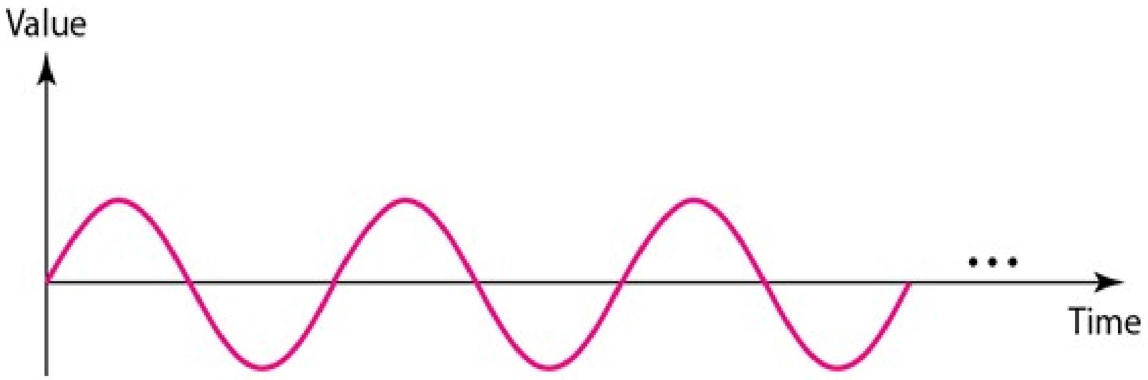
b. Digital

In data communications, we commonly use periodic analog signals and nonperiodic digital signals

Analog Signals

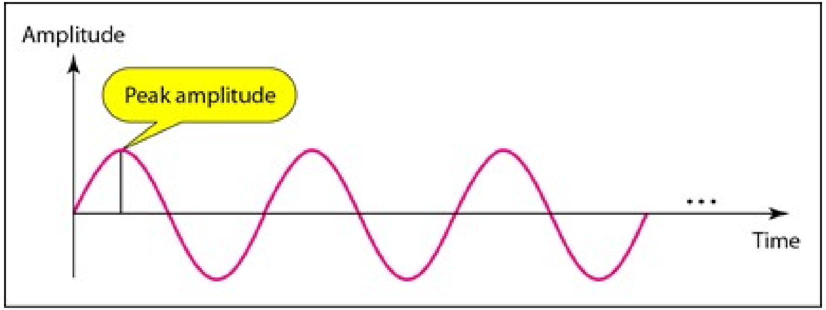
Periodic Analog Signal

- Periodic analog signal can be classified as simple or composite
- Sine Wave
  - Most fundamental form of a periodic analog signal
  - Fully described by its peak amplitude, frequency and phase

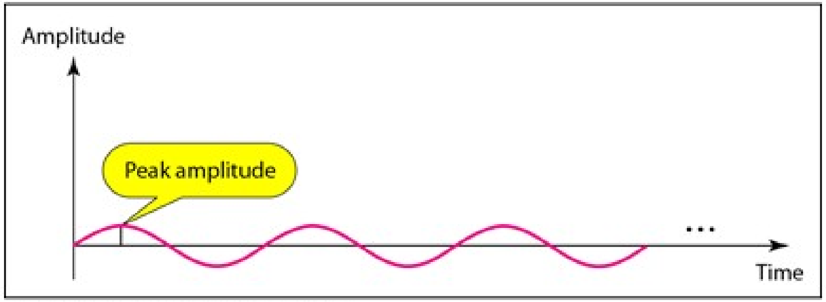


Peak Amplitude

Absolute value of a signal's highest intensity proportional to the energy it carries

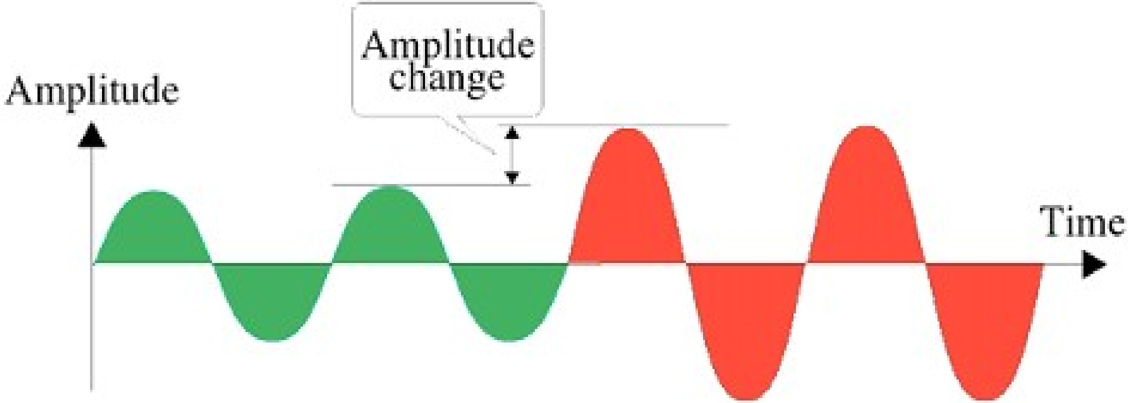


a. A signal with high peak amplitude



b. A signal with low peak amplitude

Amplitude change

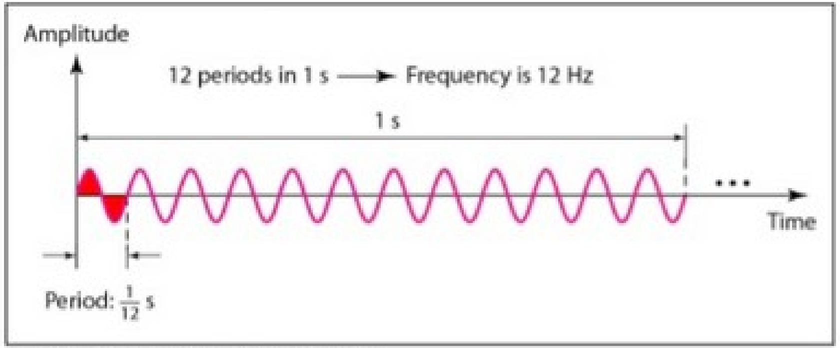


Frequency

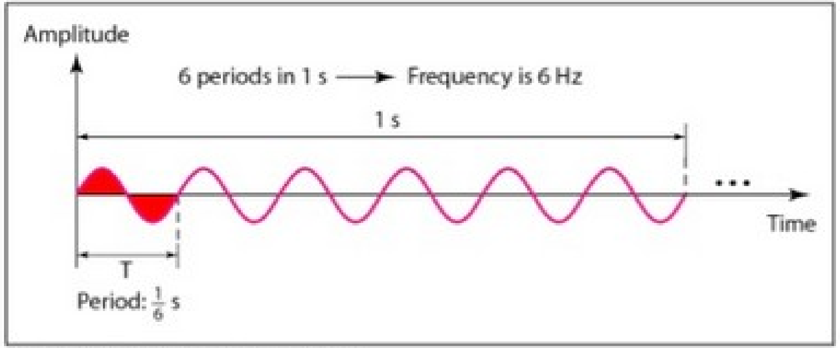
- Period (T)
  - Amount of time in seconds a signal needs to complete 1 cycle.
- Frequency (f)
  - Number of periods (or cycles) in 1 second, hertz
  - Measure of the rate of change of a signal with respect to time
- T and f are inverse of each other:

$$f = \frac{1}{T} \qquad T = \frac{1}{f}$$

Frequency



a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

Frequency

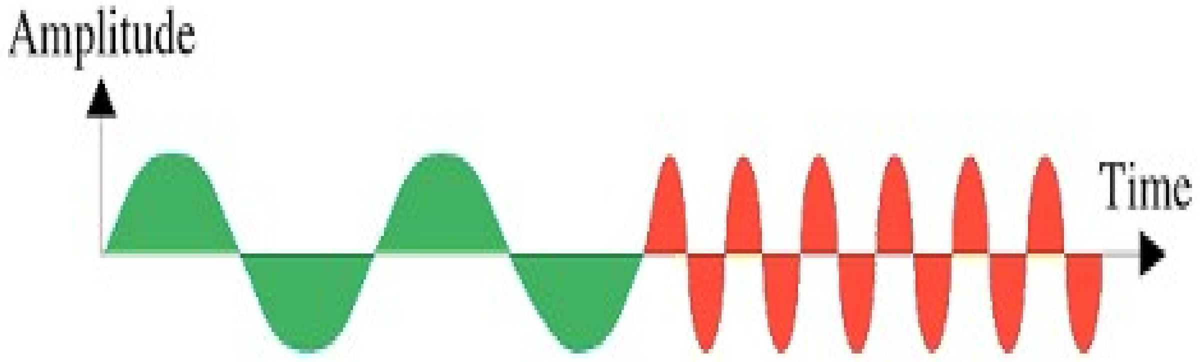
Period		*Frequency*	
Unit	*Equivalent*	*Unit*	*Equivalent*
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^{3} Hz
Microseconds (μs)	10^{-6} s	Megahertz (MHz)	10^{6} Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^{9} Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Frequency

- Change in a short span of time means high frequency
- Change over a long span of time means low frequency

- If a signal does not change at all, its frequency is zero
- If a signal changes instantaneously, its frequency is infinite

Frequency change

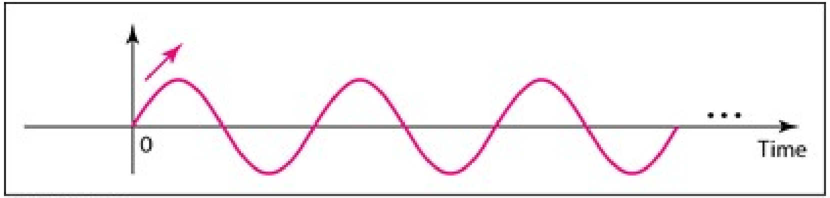


Phase

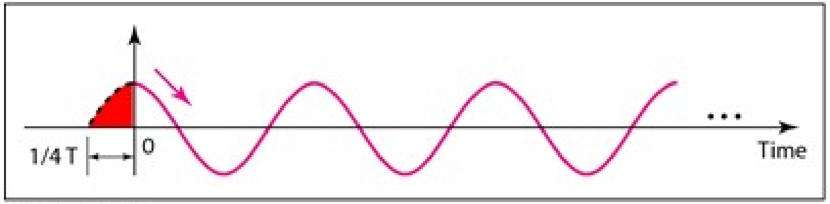
- Position of waveform relative to time 0
- The amount of shift in the time axis (x-axis)
- Measured in degrees or radians

$$360\,deg = 2\,\pi\ rad$$

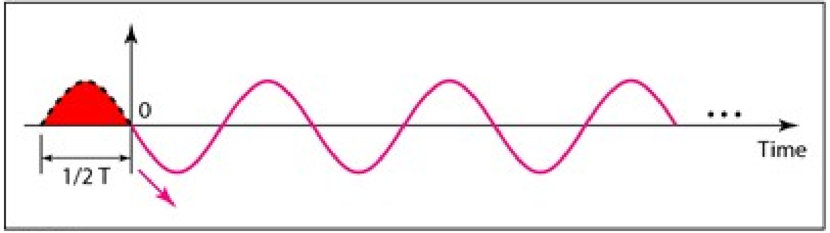
Phase



a. 0 degrees

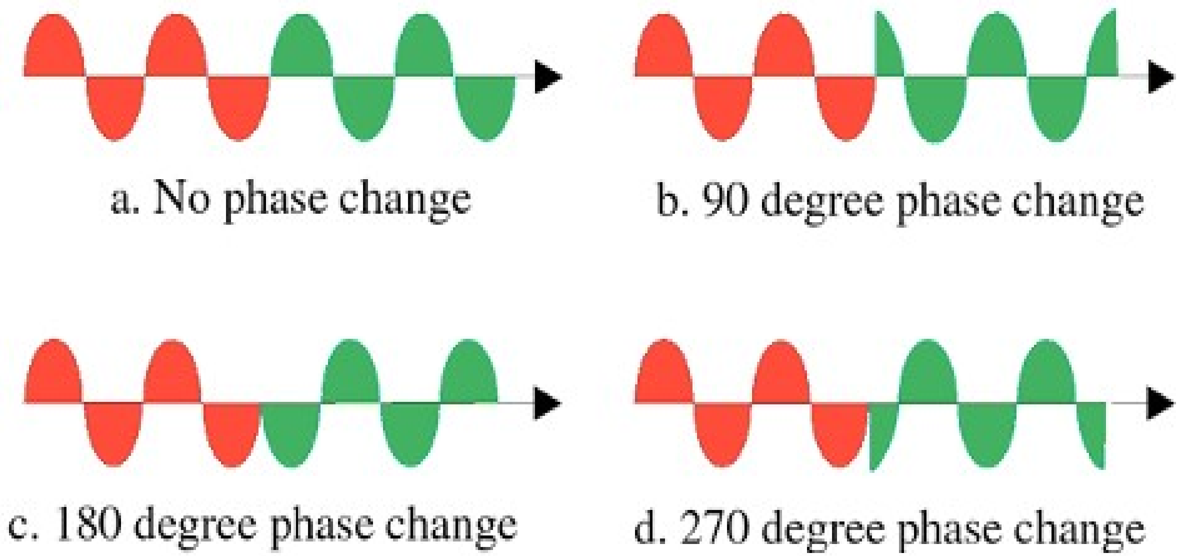


b. 90 degrees



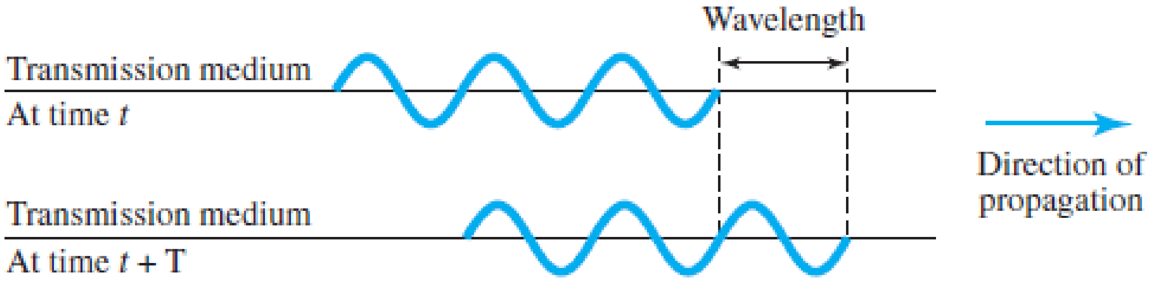
c. 180 degrees

Phase change



Wavelength

- Distance a simple signal can travel in one period (λ)
- Binds T or f of a simple sine wave to the propagation speed of the medium
- $\lambda = cT = \frac{c}{f}$ where c = propagation of light: 3 x 10^8 m/s
- Unit is micrometers (microns)

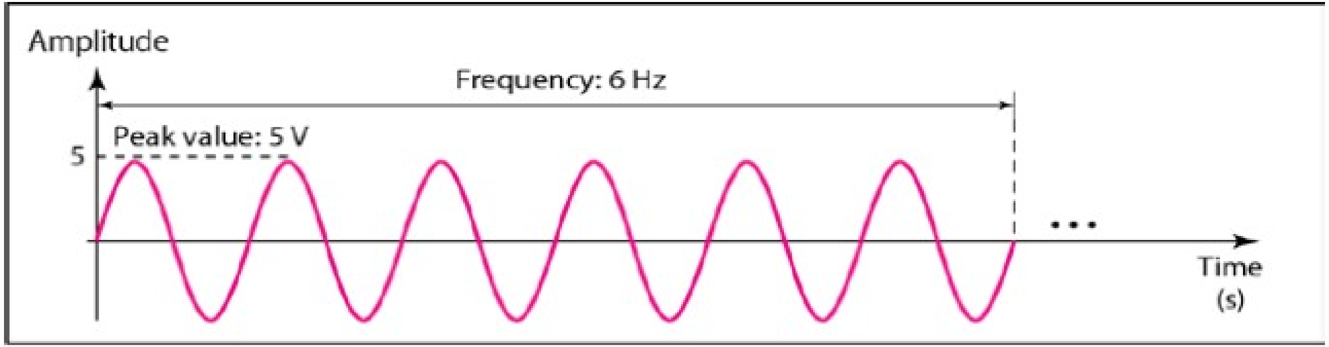


Time and Frequency Domain

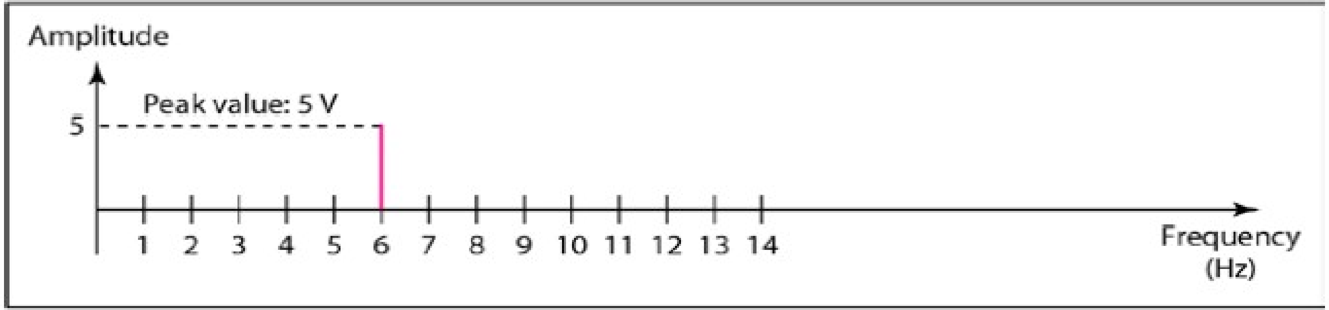
- Time-domain plot shows changes in signal amplitude with respect to time
- Frequency-domain plot is concerned with only the peak value and the frequency

- A complete sine wave in the time domain can be represented by one single spike in the frequency domain
- The position of the spike shows the frequency; its height shows the peak amplitude

Time and Frequency Domain

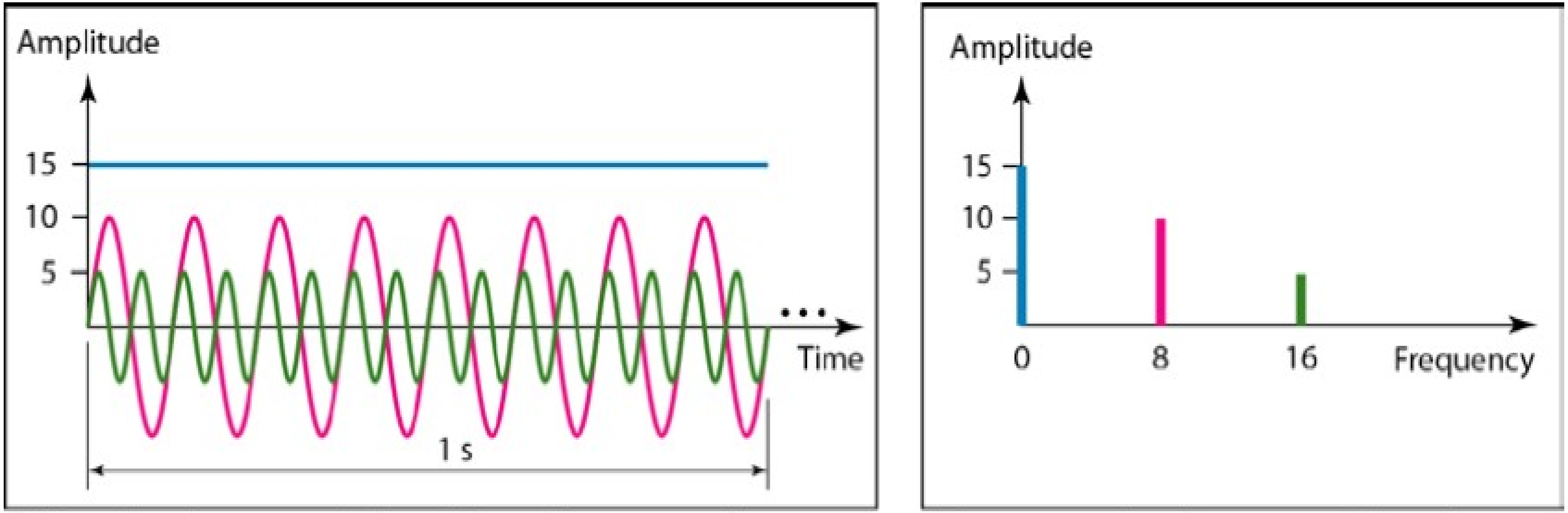


a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

Time and Frequency Domain



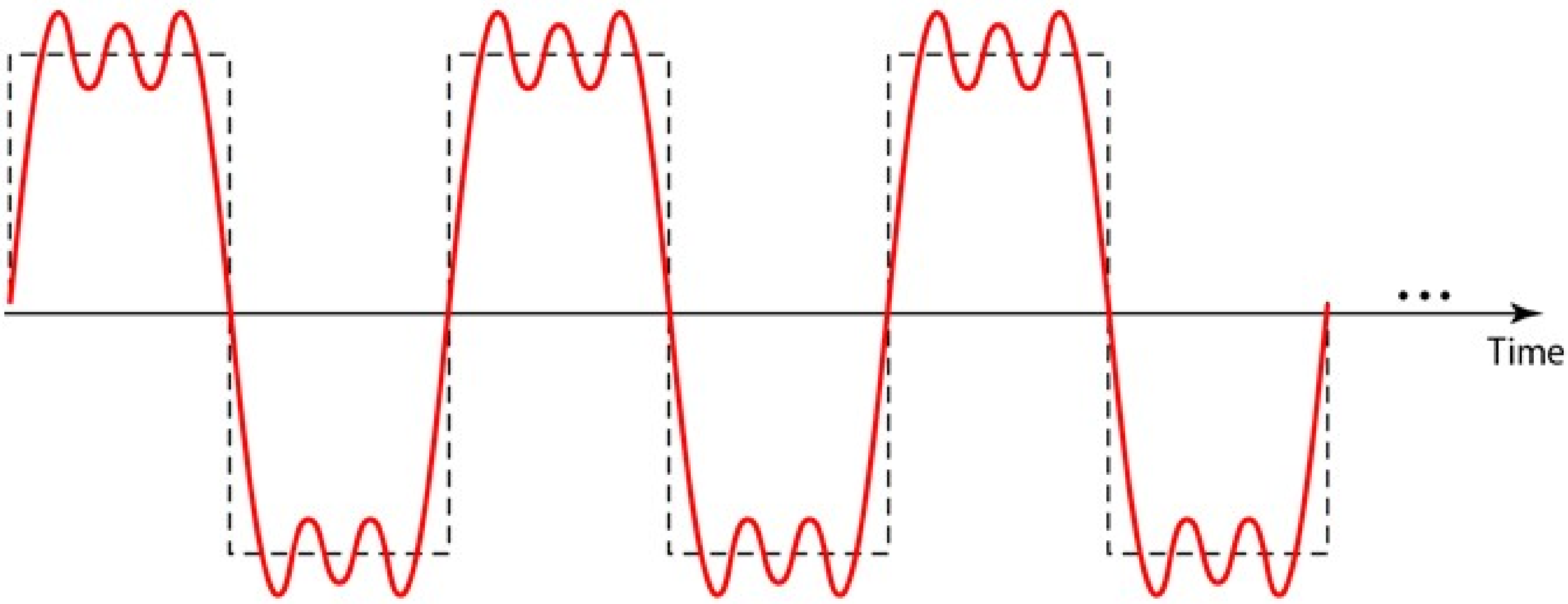
a. Time-domain representation of three sine waves with frequencies 0, 8, and 16

b. Frequency-domain representation of the same three signals

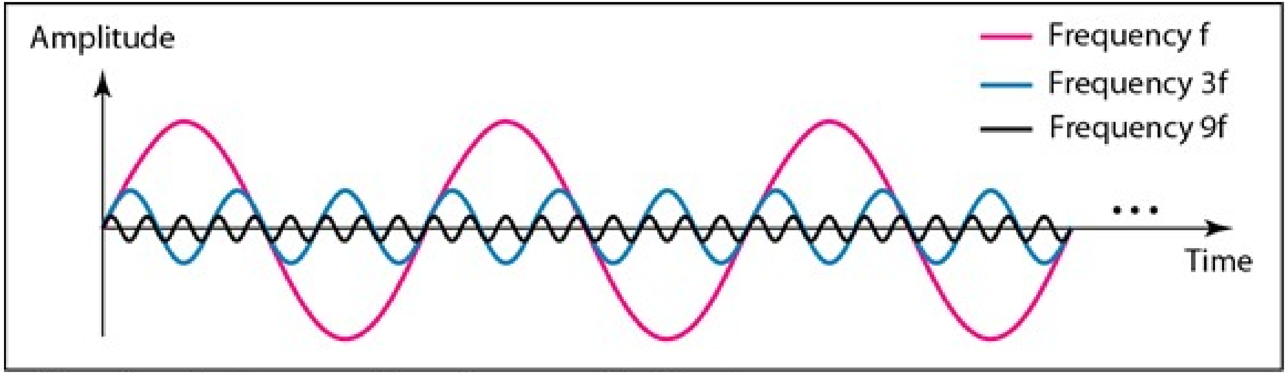
Composite Signals

- Made up of many simple sine waves
- Jean-Baptiste Fourier - any composite signal is actually a combination of simple sine waves with different *frequencies*, *amplitude*, and *phases*
- If the composite signal is periodic, the decomposition gives a *series of signals with discrete frequencies*
- If the composite signal is nonperiodic, the decomposition gives a *combination of sine waves with continuous frequencies*

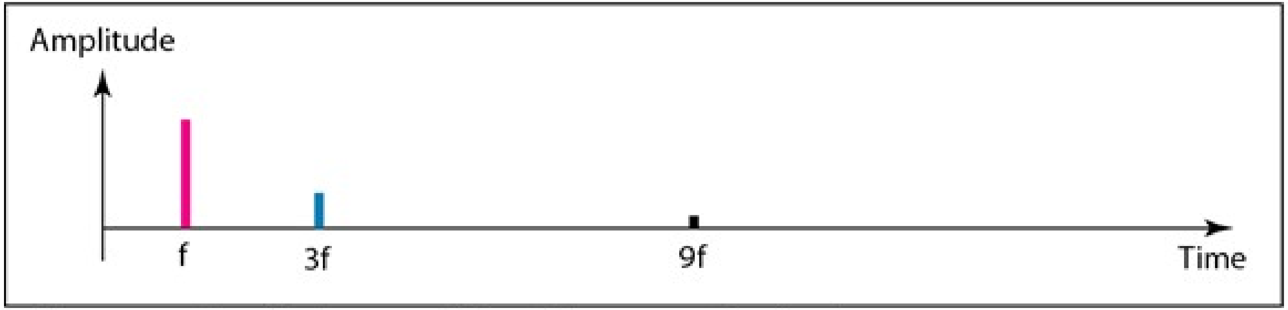
Periodic Composite Signal



Periodic Composite Signal

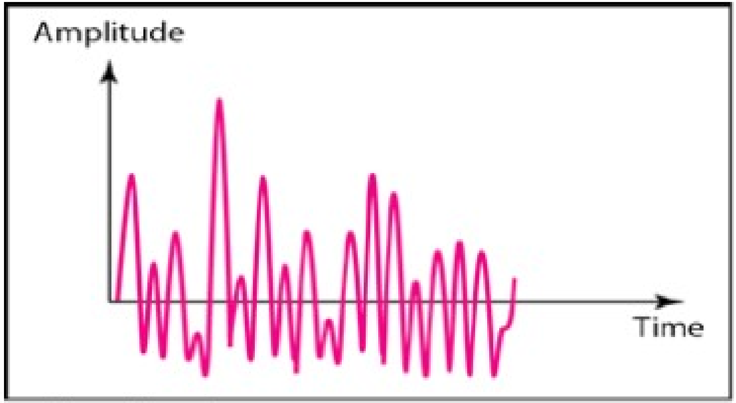


a. Time-domain decomposition of a composite signal

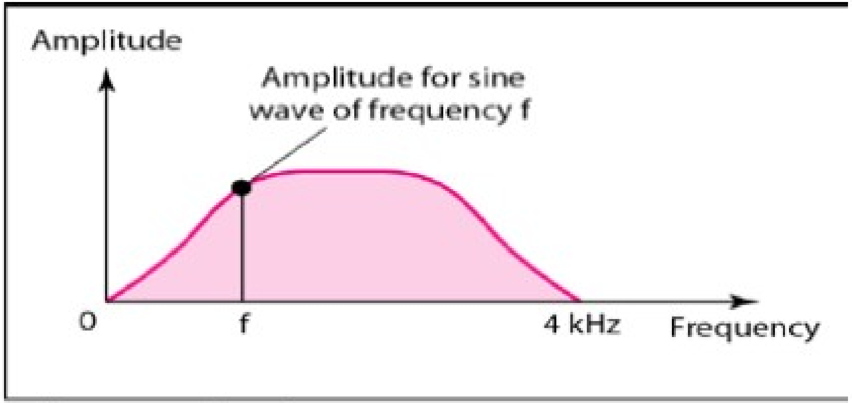


b. Frequency-domain decomposition of the composite signal

Nonperiodic Composite Signal



a. Time domain

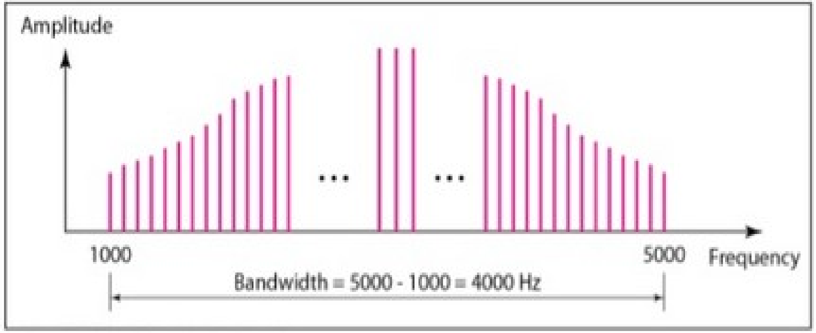


b. Frequency domain

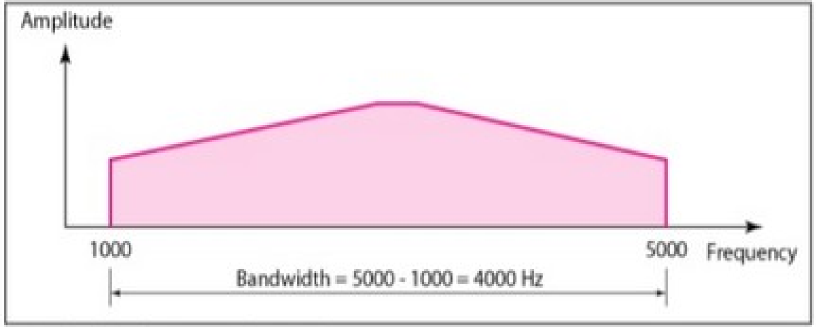
Bandwidth

- Range of frequencies contained in a composite signal
- Ex. Given a composite signal with frequencies between 1000Hz and 5000Hz

$$\textit{Bandwidth} = 5000 - 1000$$
$$= 4000\text{Hz}$$



a. Bandwidth of a periodic signal



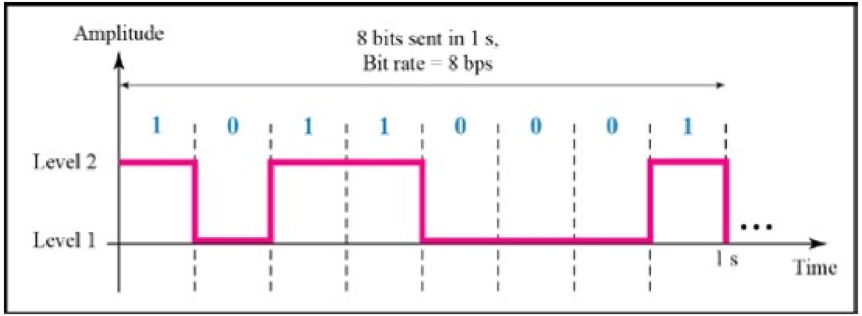
b. Bandwidth of a nonperiodic signal

Digital Signal

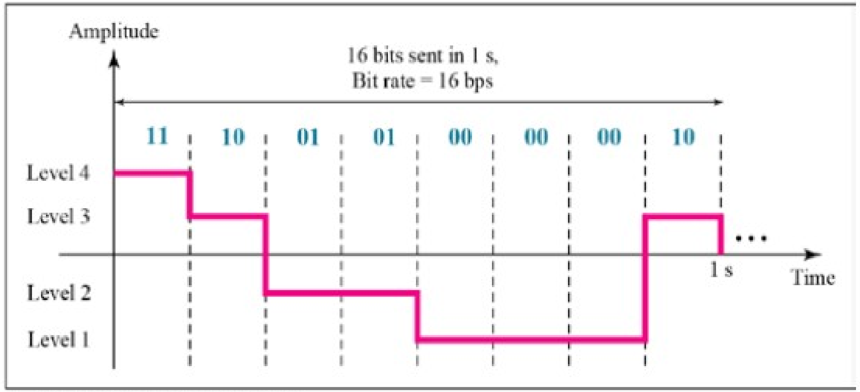
Digital Signal

- A "1" can be encoded as a positive voltage and a "0" as zero voltage
- A digital signal can have more than two levels
  - We can send more than 1 bit for each level
- If a signal has L levels, each level needs $\log_2 L$ bits
  - Ex. if digital signal has eight levels

    $\log_2 8 = 3$



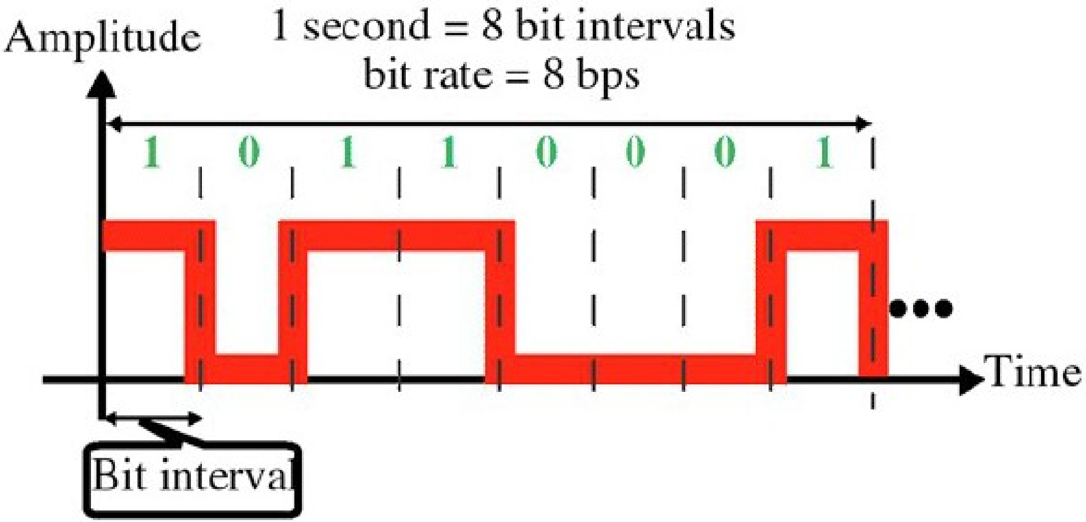
a. A digital signal with two levels



b. A digital signal with four levels

Bit Rate

- Most digital signals are nonperiodic
- Bit rate – number of bits sent in 1 second, expressed as bits per second (bps)
- Bit interval = 1 / bit rate



Bit Length

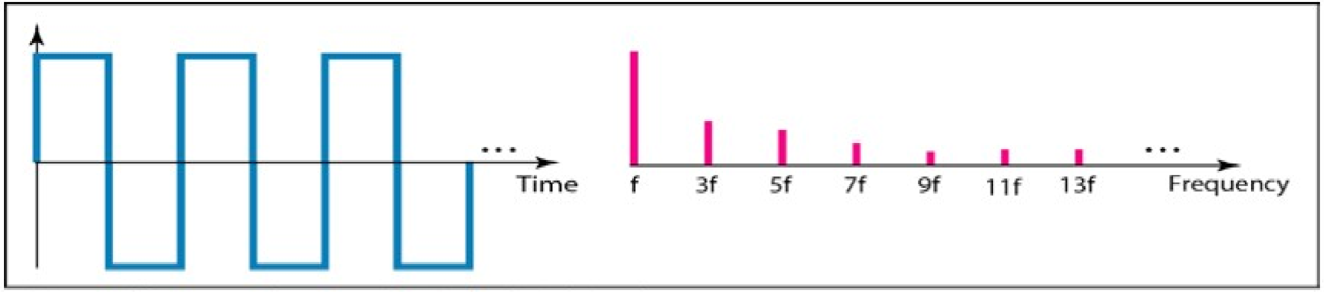
Distance one bit occupies on the transmission medium

$$Bit\ length = propagation\ speed \times bit\ duration$$

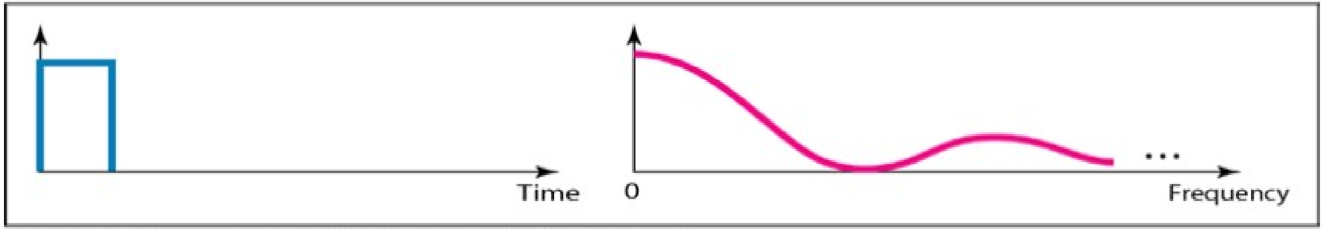
Digital Signal as a Composite Signal

- Based on Fourier analysis, a digital signal is a composite analog signal with infinite bandwidth
- In a time domain plot, it comprises connected vertical and horizontal line segments
  - Vertical line means a frequency of infinity
  - Horizontal line means a frequency of zero
- Going from frequency of zero to frequency of infinity (and vice versa) implies all frequencies in between are part of the domain

Digital Signal as a Composite Signal



a. Time and frequency domains of periodic digital signal



b. Time and frequency domains of nonperiodic digital signal

Transmission of Digital Signals

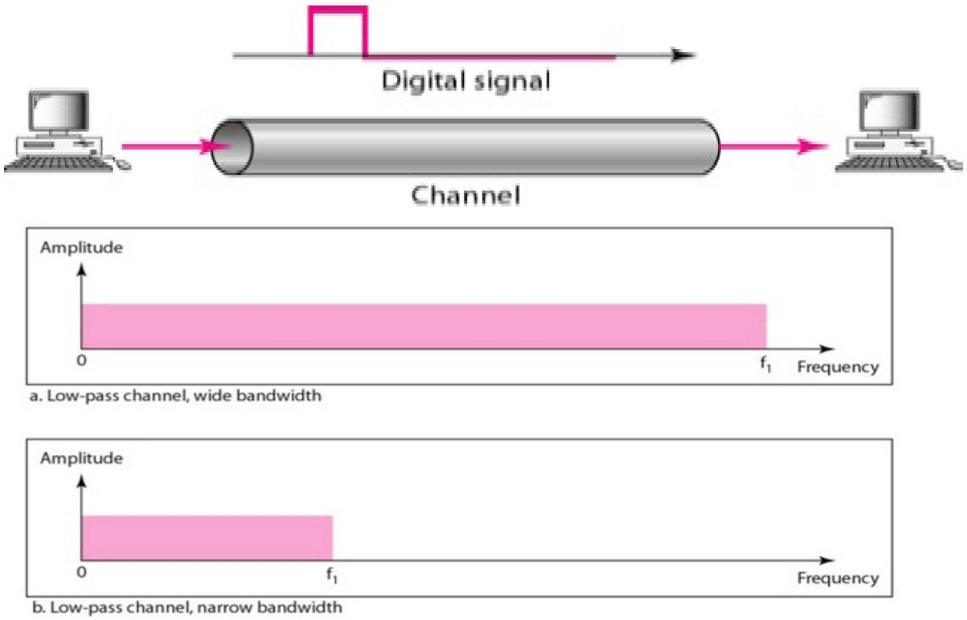
How can we send a digital signal from point A to point B?

- Data communications use nonperiodic digital signal
  - A digital signal is a composite analog signal with an infinite bandwidth
- Two approaches: baseband and broadband

Baseband Transmission

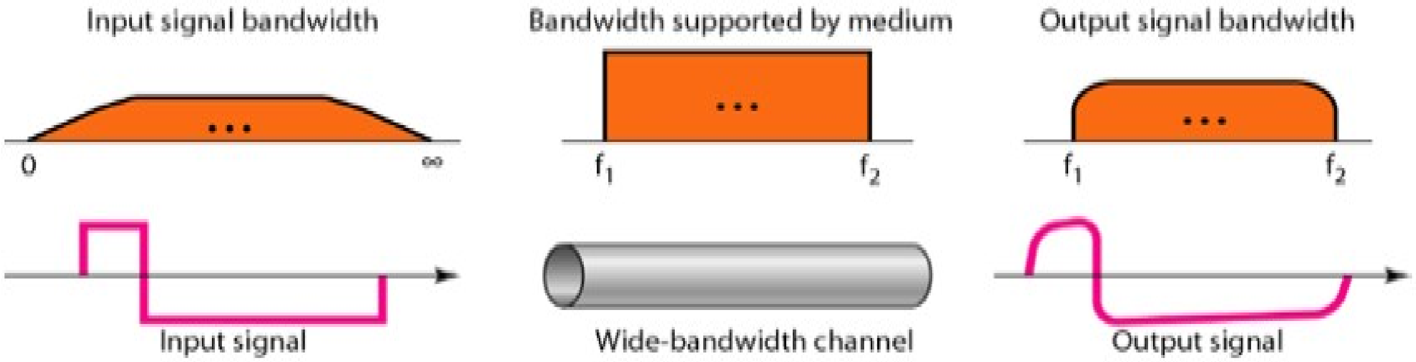
- Sends a digital signal without changing it to analog signal
- Requires a dedicated low-pass channel, a channel with a bandwidth that starts from zero

Bandwidth extending from DC to some upper frequency limit



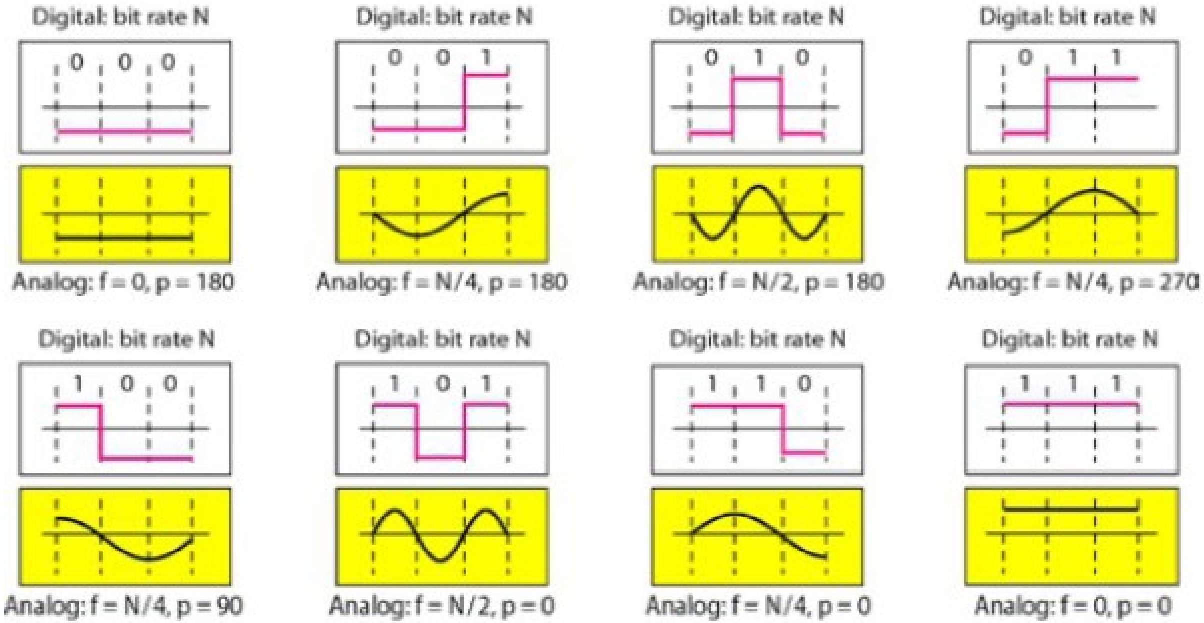
Low-Pass Channel with Wide Bandwidth

To preserves the shape of the digital signal, the entire spectrum should be sent



Low-Pass Channel with Narrow Bandwidth

- Approximate the digital signal with an analog signal
- Rough approximation

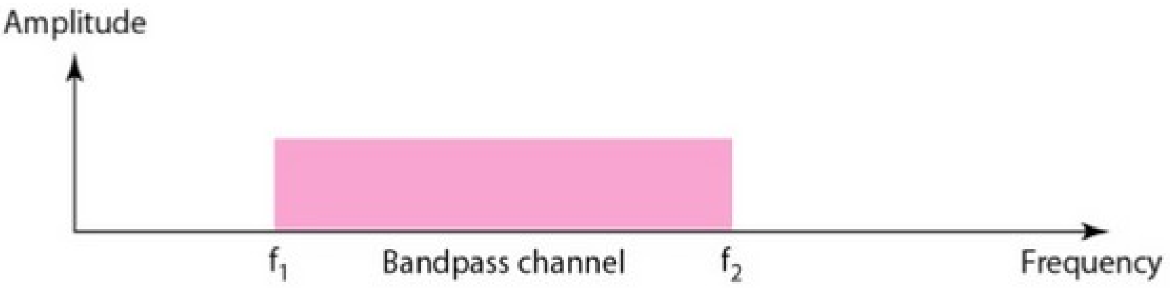


In baseband transmission, the required bandwidth is proportional to the bit rate; if we need to send bits faster, we need more bandwidth

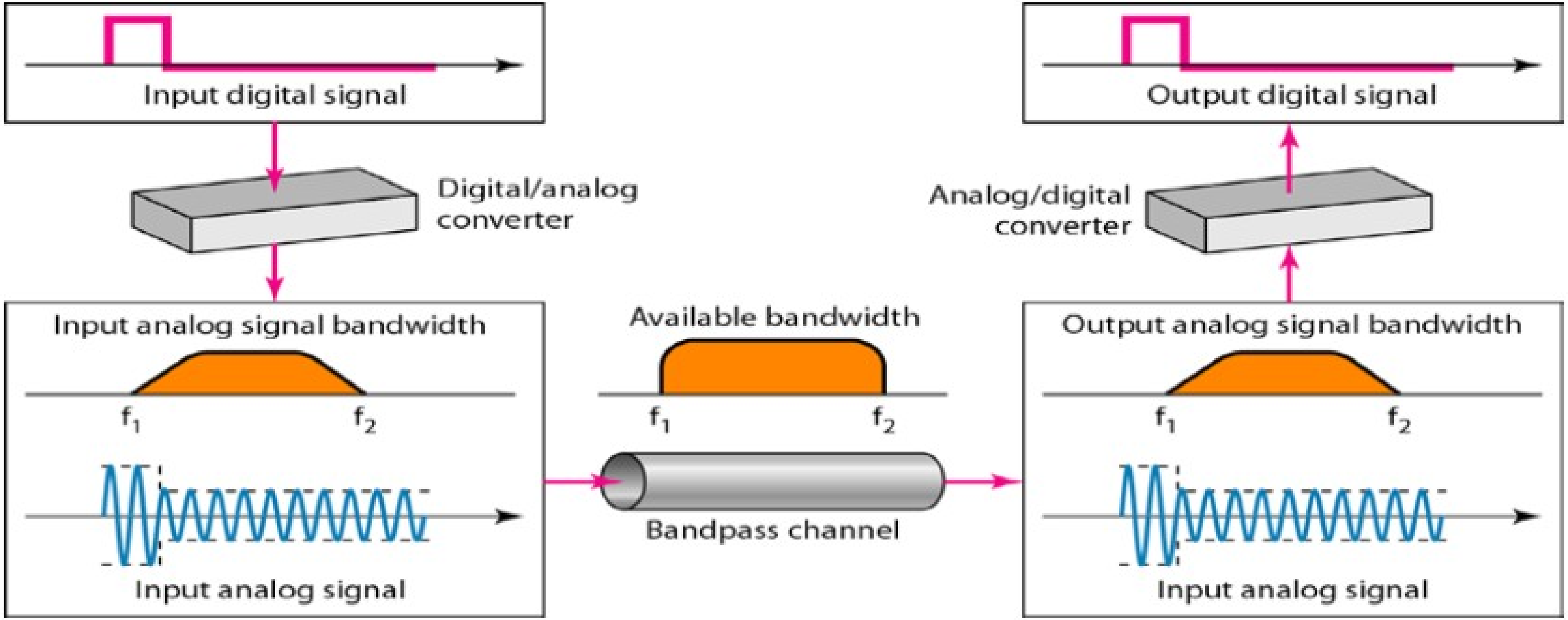
Broadband Transmission

- Changes the digital signal to analog signal for transmission (also called modulation)
- Allows the use of bandpass channel – a channel with a bandwidth that does not start from zero

Bandwidth covering a range of frequencies



Broadband Transmission



References

- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.